THE TENSOR PRODUCT DIFFERENTIAL GRADED ALGEBRA AS A TOTAL COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the comparison target is the in-source Definition definition-tensor-product (navigation stub); the degrees match since the bidegree (p, q) summand carries deg(a) = p.

Editorial note. The label lemma-total-complex-tensor-product names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Tensor product dga. Same unit, Definition definition-tensor-product. The proof checks the total-complex differential against the differential rule in Definition definition-tensor-product.

Source reference. Original reference 065W: sources/stacks-dga/pinned-original-source.tex, source lines 341–341. Prerequisite pointer invoked at proof line(s) 370; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let R be a ring. Let (A, d), (B, d) be differential graded algebras over R. Denote $A^\bullet$, $B^\bullet$ the underlying cochain complexes. As cochain complexes of R-modules we have*

$$(A \otimes_R B)^\bullet = Tot(A^\bullet \otimes_R B^\bullet).$$

Proof. Recall that the differential of the total complex is given by $\mathrm{d}_1^{p,q} + (-1)^p \mathrm{d}_2^{p,q}$ on $A^p \otimes_R B^q$. And this is exactly the same as the rule for the differential on $A \otimes_R B$ in Definition 065W. □

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